

Homework 1

ORF 245 - Fundamentals of Statistics

Due: Feb. 7th, 2025, End-of-Day (11:59pm)

Notes:

- Collaboration with other students is encouraged, but you must i) write up your own answers, and ii) acknowledge any sources or students you consulted.
 - Submit the homework on Gradescope, and be sure to assign each page to the relevant question(s).
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Exercise: 1. (Median Absolute Deviation) Consider the following data set:

$$x_1 = 10.03$$

$$x_2 = 41.61$$

$$x_3 = 5.32$$

$$x_4 = 7.78$$

$$x_5 = 8.5$$

$$x_6 = 0.2$$

$$x_7 = 1.1$$

- (a) Compute the range of $x_1, \dots, x_n$.
- (b) Calculate the sample variance and standard deviation of $x_1, \dots, x_n$.
(**Note:** In this course, we've defined the sample variance as dividing by n , and not $n - 1$.)
- (c) Another measure of variability, which is more robust than the standard deviation or variance, is called the *median absolute deviation*, which we'll denote by $\text{MAD}(x_1, \dots, x_n)$. For a data set $x_1, \dots, x_n$, the sample MAD is defined as:

$$\text{MAD}(x_1, \dots, x_n) = \text{median}\{ |x_i - x_{\text{med}}|, \ i = 1, \dots, n \}.$$

That is, MAD is the median of each data point's absolute deviation from the sample median.

Compute MAD of $x_1, \dots, x_n$ above.

(d) Let $z_1, \dots, z_n$ be the following slightly different data set:

$$z_1 = \mathbf{100.3}$$

$$z_2 = 41.61$$

$$z_3 = 5.32$$

$$z_4 = 7.78$$

$$z_5 = 8.5$$

$$z_6 = 0.2$$

$$z_7 = 1.1$$

Recompute the sample standard deviation and MAD of $z_1, \dots, z_n$. Compare these values with your answers from parts (b) and (c), what changes do you see?

Exercise: 2. (Additive Properties) Suppose we have two data sets $x = \{x_1, \dots, x_n\}$ and $y = \{y_1, \dots, y_n\}$ and let $z = x + y = \{x_1 + y_1, \dots, x_n + y_n\}$. Recall the definitions of sample mean and variance:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \qquad \sigma_x^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2.$$

(a) Prove or disprove that the mean satisfies the additive property, i.e., prove or disprove that $\bar{z} = \bar{x} + \bar{y}$.

(**Note:** to disprove the statement, it is only necessary to give a counterexample. I.e. come up with two specific datasets x and y , of any numbers and any size, for which this equality is not satisfied. The same principle applies to other “prove or disprove” problems.)

(b) Prove or disprove that the median satisfies the additive property, i.e., prove or disprove that $z_{\text{med}} = x_{\text{med}} + y_{\text{med}}$.

(c) Give an example of two specific data sets $x_1, \dots, x_n$ and $y_1, \dots, y_n$ (of any numbers and any size) showing that variance does **not** satisfy the additive property. That is, disprove $\sigma_z^2 = \sigma_x^2 + \sigma_y^2$.

Exercise: 3. (Equality of Sets) Let A , B , and C be three sets, all of which are subsets of some S .

(**Note:** You can absolutely disprove a statement using visual or numerical counterexamples, but specific examples (whether drawings or numerical) are not enough to prove a statement.)

- (a) Prove: $A \cup B = (A \cap B) \cup (A \cap B^C) \cup (A^C \cap B)$
- (b) Prove or disprove: $A \cup (B \cap C) = (A \cap B) \cup (A \cap C)$
- (c) Prove or disprove: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

Exercise: 4. (Working with Probability Axioms) Suppose we have an unfair die where the sample space is given by $S = \{1, 2, 3, 4, 5, 6\}$, and the probabilities of each outcome are:

$$\begin{aligned}\mathbb{P}(\{1\}) &= \mathbb{P}(\{2\}) = \mathbb{P}(\{3\}) = \frac{1}{4}, \\ \mathbb{P}(\{4\}) &= \mathbb{P}(\{5\}) = \mathbb{P}(\{6\}) = \frac{1}{12}.\end{aligned}$$

- (a) Let $A = \{1, 2, 4\}$. Compute $\mathbb{P}(A)$ and $\mathbb{P}(A^C)$.
- (b) Let $B = \{\text{the roll is an odd number}\}$. Compute $\mathbb{P}(B)$.
- (c) Let $C = \{\text{the roll is a prime number}\}$. Compute $\mathbb{P}(C)$.
- (d) Compute $\mathbb{P}(A \cap B)$, $\mathbb{P}(A \cup B)$, and $\mathbb{P}(A \cap C)$.

Exercise: 5. [OPTIONAL] (Properties of Probability)

Note: This is an *optional* exercise – it will not be graded, will not factor into your grade (either positively or negatively), but we will release solutions after the submission deadline.

Let S be the sample space of some experiment, and $\mathbb{P}$ a probability measure. Let $A \subset S$, and $B \subset S$ be any two events. Use the Axioms of Probability to prove the following properties:

- (a) $\mathbb{P}(A^C) = 1 - \mathbb{P}(A)$.
- (b) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.
- (c) $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$.

(*Hint: You can assume that part (b) has already been proved.*)

- (d) (**Challenging**) If $B_1, \dots, B_n$ are mutually exclusive and collectively exhaustive, then $\mathbb{P}(A) = \sum_{i=1}^n \mathbb{P}(A \cap B_i)$.

(*Hint: First, try to show that $A = \cup_{i=1}^n (A \cap B_i)$. Then, show that the sets $A \cap B_1, \dots, A \cap B_n$ are mutually exclusive, and conclude using Axiom 3 of Probability.*)